

Aprendizaje de Máquina

U1: Introducción

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Section 1

Proximity Functions on Sets

Between Subsets

The concept of **proximity** can be extended to subsets of X . Let U be a set containing subsets of X . That is, $D_i \subset X$, $i = 1, \dots, k$ and $U = \{D_1, \dots, D_k\}$.

A **proximity measure** ϕ on U is a function

$$\phi : U \times U \rightarrow \mathbb{R}$$

that satisfied

$$\begin{aligned} \exists d_0 \in \mathbb{R} : \quad & -\infty < d_0 \leq d(D_i, D_j) < \infty, & \forall D_i, D_j \in U \\ & d(D_i, D_i) = d_0, & \forall D_i \in U \\ & d(D_i, D_j) = d(D_j, D_i), & \forall D_i, D_j \in U \end{aligned}$$

Usually, the proximity measures between two sets D_i and D_j are defined in terms of proximity measures between their elements.

Between a Point and a Set

In many clusterings schemes, a vector x is assigned to a cluster C taking into account the proximity between x and C

$$\phi(x, C)$$

There are two general directions for the definition of $\phi(x, C)$:

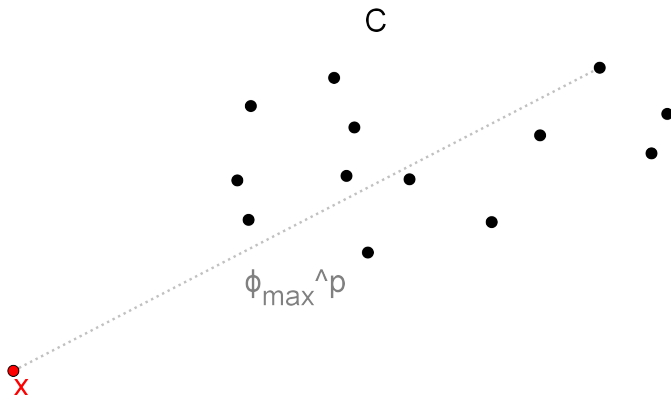
- **All the points** of C contribute to $\phi(x, C)$.
- C is equipped with a **representative point** and the proximity between x and C is measured as the proximity between x and the representative of C .

Between a Point and a Set

Approach: **All the points** contribute:

- The **max proximity function**:

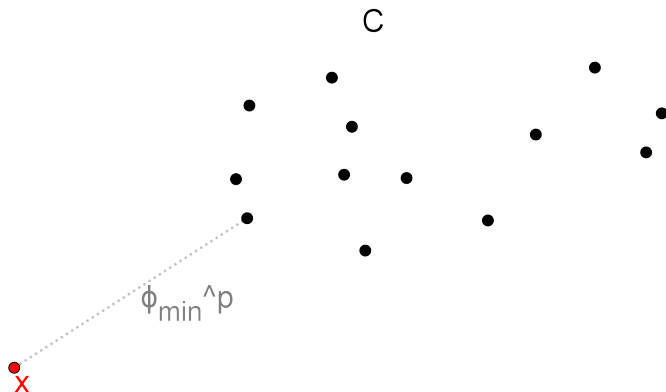
$$\phi_{max}^{ps}(x, C) = \max_{y \in C} \phi(x, y)$$



Between a Point and a Set

- The **min proximity function**:

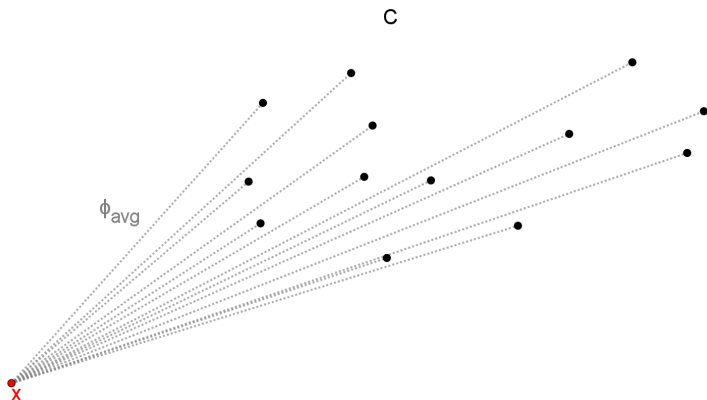
$$\phi_{min}^{ps}(x, C) = \min_{y \in C} \phi(x, y)$$



Between a Point and a Set

- The **average proximity function**:

$$\phi_{avg}^{ps}(x, C) = \frac{1}{|C|} \sum_{y \in C} \phi(x, y)$$



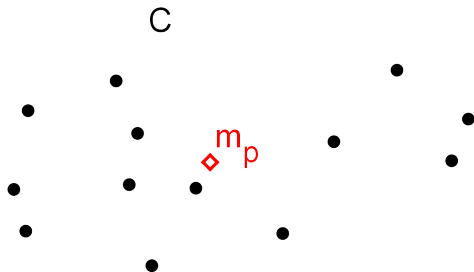
Point Representative

Typical choices for a **point representative** of a cluster are:

- The **centroid** (**mean vector** or **mean point**):

$$m_p = \frac{1}{|C|} \sum_{y \in C} y$$

It may not work well we deal with points of a discrete space $\mathbb{F}^I$, because it is possible for m_p to lie outside $\mathbb{F}^I$.



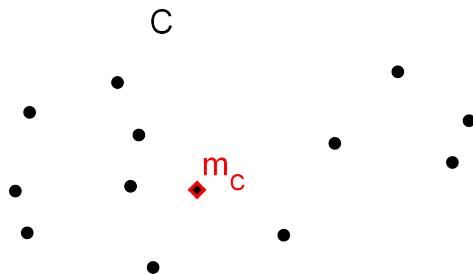
Point Representative

- The **medoid** (**mean center**) $m_c \in C$ is defined as the point for which:

$$\sum_{y \in C} d(m_c, y) \leq \sum_{y \in C} d(z, y), \quad \forall z \in C$$

where d is a dissimilarity measure between two points.

When similarity measures are involved, the inequality is reversed.



Set Representatives

Linear shaped clusters (or hyperplanar in the general case) cannot be accurately represented by a single point. In such case we use **hyperplane as representative** of the clusters.

The general equation of a hyperplane H is

$$\sum_{j=1}^l a_j x_j + a_0 = a^T x + a_0 = 0$$

where $x = (x_1, \dots, x_l)^T$ and $a = (a_1, \dots, a_l)^T$ is the weight vector of H .

In the case of Euclidean distance between two points,

$$d(x, H) = \frac{|a^T x + a_0|}{\|a\|}$$

where $\|a\| = \sqrt{\sum_{j=1}^l a_j^2}$.

Set representative

Clusters that are circular or **hyperspherical** shaped are frequently encountered in computer vision applications.

The general equation of a hypersphere Q is

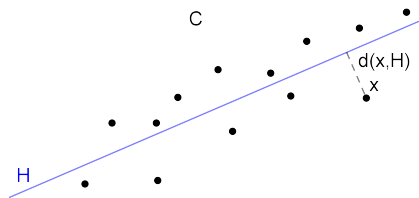
$$(x - c)^T (x - c) = r^2$$

where c is the center of the hypersphere and r its radius.

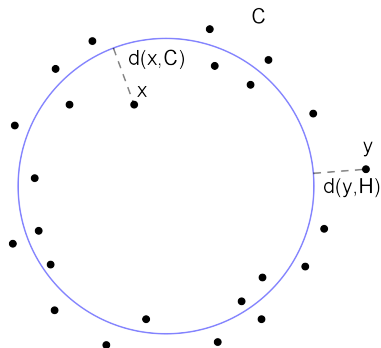
The distance from x to Q is defined as

$$d(x, Q) = \min_{z \in Q} d(x, z)$$

Set Representatives



(a) Distance from a point to a hyperplane



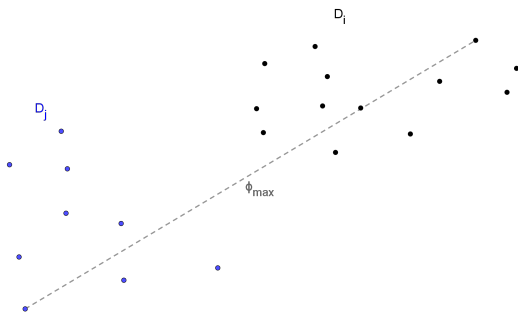
(b) Distance from a point to a hypersphere

Between two sets

- The **max proximity function**:

$$\phi_{\max}^{ss}(D_i, D_j) = \max_{x \in D_i, y \in D_j} \phi(x, y)$$

If ϕ is a similarity measure $\phi_{\max}^{ss}$ is a measure but it is not metric.

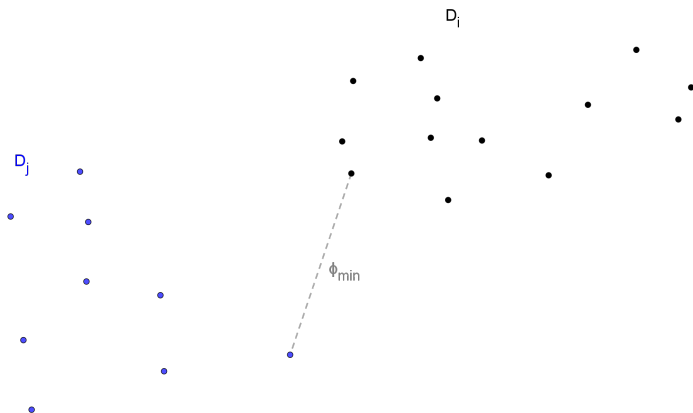


Between two sets

- The **min proximity function**:

$$\phi_{min}^{ss}(D_i, D_j) = \min_{x \in D_i, y \in D_j} \phi(x, y)$$

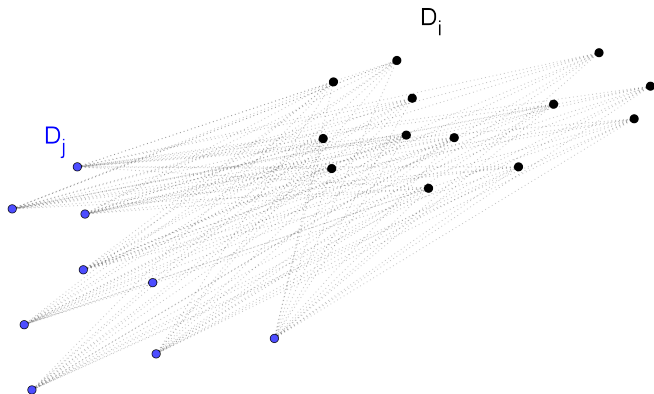
If ϕ is a dissimilarity measure, ϕ_{min}^{ss} is a measure, but is not a metric.



Between two sets

- The **average proximity function**:

$$\phi_{avg}^{ss}(D_i, D_j) = \frac{1}{|D_i||D_j|} \sum_{x \in D_i} \sum_{y \in D_j} \phi(x, y)$$



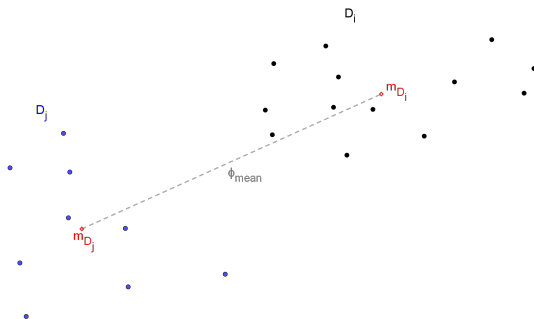
Between two sets

- The **mean proximity function**:

$$\phi_{mean}^{ss}(D_i, D_j) = \phi(m_{D_i}, m_{D_j})$$

where m_{D_i} is the **point representative** (for instance, the centroids) of D_i , $i = 1, 2$.

The mean proximity function is a measure provided that ϕ is a measure.



References

- Thoedoridis, S. & Koutroumbas, K. *Pattern Recognition*, 4th ed., Academic Press, 2009